

Magnetic Equivalent Circuit for Multistage Axial Flux Surface Permanent Magnet Machines

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ABSTRACT This article presents a magnetic equivalent circuit for the electromagnetic analysis of axial flux permanent magnet synchronous machines. The proposed methodology is based on a quasi-3D approach and incorporates the non-linear iron core behavior, the rotor motion, and magnetic leakage effects. A key strength of the proposed approach is its adaptability to different machine configurations, starting from a single-stage network and extended to multistage topologies with different stator-rotor arrangements. This article also provides a detailed guide to constructing the reluctance network, including simplifications to facilitate adoption by less experienced designers. Validation is performed against both three-dimensional (3-D) finite-element analysis simulations and experimental tests, demonstrating good agreement in key performance metrics, such as magnetic flux density, back electromotive force, and torque. The results validate the performance of the proposed magnetic equivalent circuit in reducing computational time while maintaining satisfactory accuracy, positioning it as a valuable tool for preliminary design and optimization of studies.

INDEX TERMS Analytical modeling, axial flux, electromagnetic analysis, finite-element analysis (FEA), magnetic equivalent circuit, permanent magnet synchronous machines, quasi-3D.

I. INTRODUCTION

Permanent magnet (PM) synchronous machines are among the most suitable topologies for electrical propulsion applications, including light and heavy-duty vehicles as well as electrically propelled aircraft. While PM machines are predominantly employed in their radial flux configuration, their axial flux counterpart (AFPM) has gained significant attention in recent decades due to its compact design and favorable key performance indexes for high number of poles and short axial length [1]. Their primary advantages include high specific torque and power density, as well as high efficiency. Despite challenges such as higher inertia and increased manufacturing costs, advancements in materials and technology have driven the axial flux configuration to a level of

technological maturity that has enabled its development and commercialization.

The design of electric machines requires a multiphysics approach, including electromagnetic, thermal, and mechanical analyses to accurately evaluate performance. Electromagnetic analysis, in particular, is generally performed using finite-element analysis (FEA) software. However, FEA poses challenges related to intensive computational requirements and extended runtimes. As a result, analytical techniques are often considered as a promising alternative, offering faster computation and seamless integration with other modeling tools and optimization algorithms [2], [3], [4], [5]. Among these techniques, the magnetic equivalent circuit (MEC) methodology provides a viable compromise between accuracy and

computational efficiency [6], [7], [8]. The MEC methodology discretizes the electric motor into distinct sections, each represented by a reluctance (or permeance) element to model its magnetic behavior [9], [10], [11], [12]. By applying principles analogous to electrical circuit theory, MEC can predict machine performance effectively. Additionally, the MEC approach can consider effects such as magnetic leakage, rotor motion, and different winding configurations, making it a versatile tool for generalized electromagnetic analysis.

While numerous studies in the literature address the development of MEC models for radial flux machines, research on axial flux configurations has been comparatively less explored. Moreover, the necessity to find an alternative to FEA is particularly pronounced for axial flux machines, since the three-dimensional nature of the flux paths forces to perform three-dimensional (3-D) simulations, making FEA models significantly more computational demanding. To overcome this issue, quasi-3D computation can be achieved by creating several 2-D submodels, representing the overall machine behavior [13]. This approach, commonly referred to as 2.5D modeling, can significantly reduce simulation time. However, its integration into larger multiphysics workflows remains challenging due to software compatibility issues and still relatively long solving times. Therefore, using the MEC approach for the performance prediction of AFPMs can be particularly beneficial.

A MEC modeling for an AFPM with an anisotropic rotor and fractional-slot distributed winding (DW) is reported in [14], where two different magnetic circuits are used for describing the motor behavior along the d - and q -axes. Despite cross-saturation effects are neglected, the reported results indicate acceptable agreement between the reluctance network predictions and FEA simulations.

In [15], Alipour-Sarabi et al. compare numerical and analytical methods for evaluating the performance of a rotor resolver. The analytical method is based on a 3-D reluctance network composed of several layers at different radii. The results from the MEC align well with 3-D FEA simulations. However, since the model is specifically developed for a resolver, the key aspects considered in its design and analysis may not necessarily align with the critical performance parameters required for other applications.

A more comprehensive model for AFPM machines using MECs is presented in [16]. This approach evaluates overall machine performance, including voltage, torque, and loss computations. A key innovation is the representation of rotation within the machine, where the PMs are treated as time-varying flux sources, simplifying the solution process and enhancing computational efficiency. The results show strong agreement with 3-D FEA and experimental validations. However, the limited detailed information about the MEC construction and reluctance elements restricts the replicability of the approach.

In [17], although a generalized MEC with a relatively simple network is proposed, the absence of torque measurements, an essential performance indicator, limits the completeness

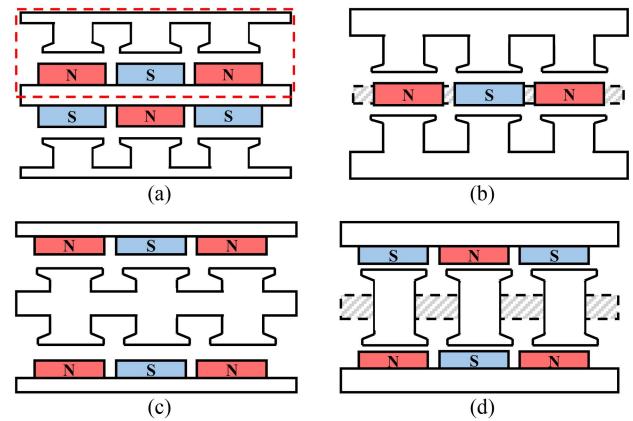


FIGURE 1. Axial flux PM machines configurations. (a) 2S1R-NS (with a single stage configuration highlighted). (b) 2S1R-NN. (c) 1S2R-NN. (d) 1S2R-NS.

of the validation. The work in [18] presents a simplified reluctance network for a yokeless AFPM; however, the validation is restricted to simulation results, with no experimental comparison. In [19] a generalized framework that can be applicable for any axial flux configuration is developed. The results show good agreement with FEA, with experimental validation yet to be conducted. Finally, Mignot et al. [20] present a magnetic circuit model that demonstrates good performance, yet the mathematical formulation is complex and remains unvalidated experimentally.

Additional studies tailored to specific configurations or applications, are also available in literature. In [21], a MEC was developed for an axial flux machine with a Halbach PM configuration, achieving high accuracy at the cost of significant network complexity and computational demand. The work in [22] proposes a reluctance network for a ring-winding machine with a detailed mathematical representation of the airgap permeance, showing good agreement with experimental results. In [23], a dedicated MEC for coreless AFPM machines was presented, yielding satisfactory results, though its applicability to machines with ferromagnetic cores remains uncertain.

In this context, this article presents a MEC for AFPM machines, designed to accommodate various machine configurations at no-load and load operating conditions, considering both distributed and concentrated windings. The proposed MEC accounts for nonlinear iron effects, rotor motion, and PM subdivision to enhance resolution and accuracy. It is based on a quasi-3D representation, allowing the user to define the number of layers to be evaluated. The model also incorporates PM leakage effects, which are particularly significant in configurations with large air gaps.

A key feature of the proposed methodology is its flexibility in modeling different axial flux machine configurations. Specifically, the different cases reported in Fig. 1 are considered: two stators and one rotor (2S1R) in north-south (NS) configuration, where the red dashed line highlights the single

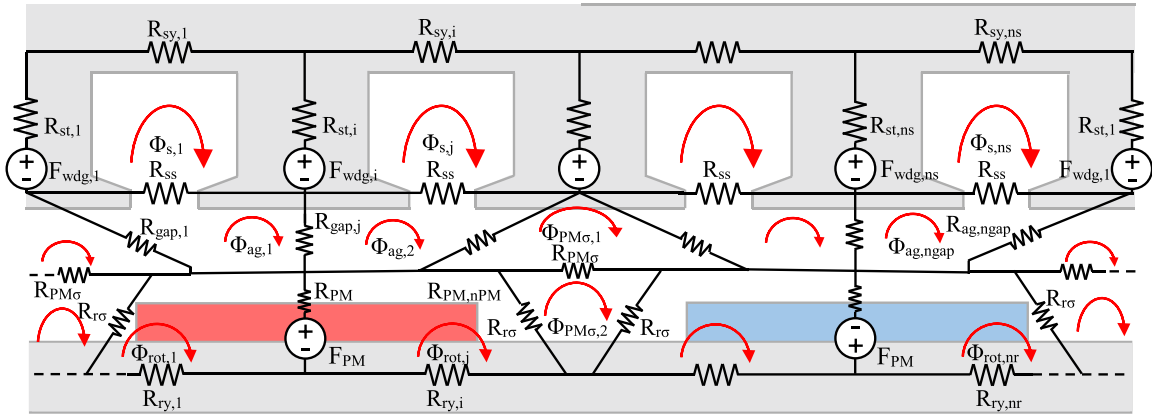


FIGURE 2. Magnetic equivalent circuit with magnetic flux meshes.

stage structure; two stators and one rotor (2S1R) in north-north (NN) configuration; one stator and two rotors (1S2R) in north-north configuration; and one stator and two rotors (1S2R) in north-south configuration. It is worth noting that the configurations shown in Fig. 1(b) and (d) are depicted with a dashed rotor and stator yoke, respectively, as they are often realized with a yokeless design due to the nature of their flux path.

This article distinguishes itself from the studies discussed above and other approaches available in the literature by proposing a flexible and simplified methodology that employs only the essential reluctance elements. The objective is to achieve a balance between the complexity of the modeled physical effects and the computational effort required, without compromising the accuracy of the results. This feature can be leveraged in design workflows, where rapid evaluation is crucial for analyzing a large number of models during the optimization phase. Additionally, experimental testing is carried out through torque and back-electromotive force measurements on an axial flux multistage surface PM machine prototype, which has not been widely reported in literature for the validation of MEC models.

II. MAGNETIC EQUIVALENT CIRCUIT

The 2-D representation of the developed MEC for a general single stage AFPM at a specific diameter D_i is shown in Fig. 2. The circuit only represents one section of the machine; however, the same topology can be repeated as many times as needed to model the entire geometry.

The total number of reluctances depends on how many distinct sections are used to discretizes the machine. Specifically, the number of stator yoke (R_{sy}) and stator tooth (R_{st}) reluctances correspond to the number of teeth in one machine periodicity (n_s). The reluctances of the air gap (R_{gap}) depend on the number of air gap segments (n_{gap}). This discretization will impact both the accuracy and the computation time.

The MEC circuit also considers leakage flux paths that have been modeled by means of slot leakage reluctances (R_{ss}) and PMs leakage reluctances ($R_{rσ}$ and $R_{PMσ}$). Indeed,

a preliminary assessment of the influence of these leakage flux paths indicates an accuracy increase of 2%–3% on the electromagnetic torque computation for the considered case studies. Furthermore, it is worth noting that incorporating these additional reluctances into the network has a negligible impact on the computational cost as their number is small compared to the total airgap reluctances.

In the proposed model, no specific reluctance element is assigned to the stator shoe region. This simplification was adopted to facilitate integration into design workflows and to minimize computational time by reducing the network complexity. In any case, the effects of saturation in the stator shoe can be evaluated in subsequent stages of the design process.

The proposed circuit is a mesh-based model, meaning that Kirchhoff's voltage law is applied to determine the unknown magnetic fluxes in each loop of the circuit, on the basis of the specified magnetomotive forces (MMFs) and the magnetic reluctances [24]. Since the reluctances representing parts of the machine composed of ferromagnetic material are nonlinear circuit elements, a system of nonlinear equations must be formulated and numerically solved to evaluate the machine performance.

To solve the MEC, each reluctance in the circuit must be computed. The general formulation for calculating magnetic reluctance is shown in (1), where l is the length of the considered magnetic flux tube, S is its cross-section, μ_r is the relative magnetic permeability, μ_0 is the vacuum permeability. The equation simplifies for prismatic or cylindrical volumes where the cross-sectional area and the relative magnetic permeability remain constant along the path length l

$$R_m = \int \frac{dl}{\mu_0 \cdot \mu_r(l) \cdot S(l)} \Rightarrow R_m = \frac{l}{\mu_0 \cdot \mu_r \cdot S} \text{ if } \begin{cases} \frac{dS}{dl} = 0 \\ \frac{d\mu_r}{dl} = 0 \end{cases} \quad (1)$$

The following formulations of the reluctances are based on the dimensions reported in Fig. 3 for the considered diameter D_i .

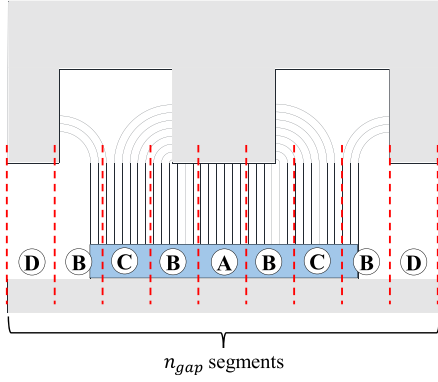


FIGURE 3. Air gap subdivision principle.

A. STATOR AND ROTOR IRON RELUCTANCES (R_{st} , R_{sy} , R_{ry})

The j th stator tooth reluctance can be computed as in (2), where h_{ts} is the tooth height, $w_{t,i}$ is the tooth width at the diameter D_i , $\mu_{r,st,j}$ is the relative permeability of the flux tube iron, $L_{core,s}$ is the radial length of the stator core and h_{ys} is the yoke thickness.

$$R_{st,j} = \frac{h_{ts}}{\mu_0 \cdot \mu_{r,st,j} \cdot L_{core,s} \cdot w_{t,i}}. \quad (2)$$

In (2), $\mu_{r,st,j}$ is the relative permeability of the iron, which is function of the magnetic flux passing through the reluctance $R_{st,j}$ as defined as

$$\mu_{r,st,j} = f(\Phi_{s,j-1} - \Phi_{s,j}). \quad (3)$$

Similarly, the j th stator yoke reluctance and the j th rotor yoke reluctance can be calculated using (4) and (5), respectively. Here, N_c is the number of stator slots, $\mu_{r,sy,j}$ and $\mu_{r,ry,j}$ represent the relative permeability of the j -th flux tube iron in the stator yoke and rotor yoke, respectively, h_{ys} and h_{yr} denote the stator and rotor yoke thicknesses, respectively, p is the number of pole pairs and $L_{core,r}$ is the radial length of the rotor iron core

$$R_{sy,j} = \frac{\pi \cdot D_i / N_c}{\mu_0 \cdot \mu_{r,sy,j} \cdot L_{core,s} \cdot h_{ys}} \quad (4)$$

$$R_{ry,j} = \frac{\pi \cdot D_i / (2p)}{\mu_0 \cdot \mu_{r,ry} \cdot L_{core,r} \cdot h_{yr}}. \quad (5)$$

In this case, the flux passing through the reluctances directly corresponds to the loop flux. Therefore,

$$\mu_{r,sy,j} = f(\Phi_{s,j}), \quad \mu_{r,ry,j} = f(\Phi_{rot,j}). \quad (6)$$

B. SLOT LEAKAGE RELUCTANCE (R_{ss})

To represent the effect of slot leakage flux, several formulations are available in the literature, many of which show good agreement with experimental validation. This work employs the specific permeance expressions developed in [25] and [26] to compute the slot leakage reluctance for DW and fractional-slot concentrated winding (FSCW), respectively. Accordingly, the following equation illustrates the formulation used to

compute the slot permeance for both winding configurations

$$\begin{cases} \lambda_{\sigma s} = \frac{h_{us}}{w_{t2}} \left[\frac{\left(\frac{w_{t1}}{w_{t2}}\right)^2 - \frac{w_{t1}^4}{4w_{t2}^4} - \ln\left(\frac{w_{t1}}{w_{t2}}\right) - \frac{3}{4}}{\left(1 - \frac{w_{t1}}{w_{t2}}\right)\left(1 - \left(\frac{w_{t1}}{w_{t2}}\right)^2\right)^2} \right], \text{ For DW} \\ \lambda_{\sigma s} = k_1 \left(\frac{h_{us} - h'_{ll}}{3w_s} \right) + k_2 K + \frac{h'_{ll}}{4w_s}, \text{ For FSCW} \end{cases} \quad (7)$$

In (7), w_{s1} and w_{s2} are the inner and outer tooth width. While h'_{ll} is the distance between two winding layers, k_1 and k_2 are the coefficients that consider the number of winding layers, calculated as in [26], while K is defined as follows:

$$K = \frac{h_a}{w_s} + \frac{h_{es}}{w_{so}} + \frac{h_{ls}}{w_s - w_{so}} \ln \frac{w_s}{w_{so}}. \quad (8)$$

C. AIR GAP RELUCTANCE (R_{gap})

The relative motion between the rotor and stator teeth leads to a continuous variation in the MEC topology at each operating point. To accurately model this behavior in the reluctance network, an adequate reluctance model should be considered. This article categorizes four different operating scenarios depicted in Fig. 3.

The number of sections in which the air gap can be subdivided is a parameter that can be chosen by the user, and it will determine the accuracy and the computation time of the algorithm. According to Fig. 3, case A corresponds to the condition where the air gap section completely faces both the stator tooth and the PM. Then, another possibility, case B, occurs when the air gap section faces the PM and only a portion of the stator tooth, resulting in an increased reluctance value. Case C considers that the air gap section is facing the PM and the stator slot entirely. Finally, Case D represents the situation where the air gap section does not encompass any portion of PM, making the magnetic flux through these sections negligible. The detailed formulation for the first three cases is presented as

$$R_{gap,j} = \begin{cases} R_{gap,min} \\ k_{gap} R_{gap,min} + (1 - k_{gap}) R_{fringe} \\ R_{fringe} \end{cases} \quad (9)$$

$$R_{gap,min} = \frac{h_g}{\mu_0 S_{gap}} \quad (10)$$

$$R_{fringe} = \frac{1}{\mu_0 S_{gap}} \left(h_g + \frac{\pi}{4} l_{fringe} \right). \quad (11)$$

Where $R_{gap,min}$ is the minimum air gap reluctance, k_{gap} defines the portion of an air gap section that faces a stator tooth, h_g refers to the air gap thickness, and S_{gap} is the air gap section area. R_{fringe} is the fringe flux reluctance defined as an additional flux path of the length of a quarter of a circumference with a radius of l_{fringe} defined as the distance between a tooth edge and the air gap section of interest.

D. PERMANENT MAGNET RELUCTANCE (R_{pm})

Each PM segment is represented as a magnetomotive force source connected in series with a magnetic reluctance. The

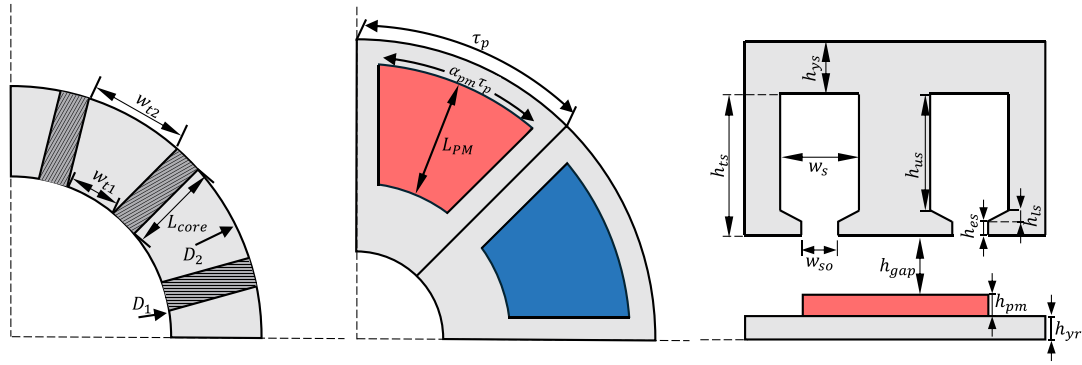


FIGURE 4. Main radial and axial dimensions of the AFSM.

amplitude of this MMF can be calculated as in (12), where B_r is the remanent flux density at the operating temperature, h_{PM} is the magnet axial thickness and $\mu_{r,PM}$ is the relative permeability of the PM material

$$F_{PM} = \frac{B_r \cdot h_{PM}}{\mu_0 \cdot \mu_{r,PM}}. \quad (12)$$

The reluctance of each PM segment can be computed as

$$R_{PM,i} = \frac{h_{PM}}{\mu_0 \cdot \mu_{PM} \cdot \tau_p \alpha_{PM} L_{PM}} \quad (13)$$

where τ_p is the pole pitch, α_{PM} is the pole-to-magnet ratio, and L_{PM} is the magnet length in the radial direction (see Fig. 4).

E. ROTOR LEAKAGE RELUCTANCES ($R_{r\sigma}$, $R_{PM\sigma}$)

The developed MEC also accounts for flux leakage occurring in the region between two consecutive magnets. Specifically, this includes leakage flux passing through the air gap between two adjacent PMs and flux traveling between a PM and the rotor yoke without linking to the stator iron, as explained in [27]. In order to account for the leakage flux caused by these phenomena, two types of reluctances are included in the MEC between the PMs. The authors in [27] developed the formulations for these reluctances, considering no saturation in the iron and a negligible magnetic field intensity produced by the armature windings, reaching a good agreement with simulation results. These magnetic reluctances can be computed as

$$R_{r\sigma} = \left[\mu_0 \cdot \frac{L_{core}}{\pi} \cdot \ln \left(1 + \frac{\pi \cdot h'_{gap}}{h_{pm}} \right) \right]^{-1} \quad (14)$$

$$R_{PM\sigma} = \left[\mu_0 \cdot \frac{L_{core}}{\pi} \cdot \ln \left(1 + \frac{\pi \cdot h'_{gap}}{\tau_p (1 - \alpha_{PM})} \right) \right]^{-1} \quad (15)$$

where, $R_{r\sigma}$ is the PM-rotor leakage reluctance, $R_{PM\sigma}$ is the PM-PM leakage reluctance, and h'_{gap} is the effective air gap that considers the slotting effect.

F. MULTISTAGE CONFIGURATIONS

The MEC depicted in Fig. 2 has been developed for the single-stage configuration. However, the circuit topology can

be readily adapted to the different multi-stage machine alternatives of Fig. 1. Adapting the circuit to these configurations requires minor modifications to the baseline case. Fig. 5(a) illustrates the modified circuit topology for the 2S1R-NN configuration, where the rotor yoke reluctances are eliminated, as the PM-to-rotor leakage elements can be neglected, assuming that the PM holder is made of nonferromagnetic materials. Fig. 5(b) presents the circuit for the 1S2R-NS configuration, where only the stator yoke reluctances disappear compared to the baseline circuit. For the 2S1R-NS and the 1S2R-NN configurations, the performance can be directly obtained by doubling the values from the baseline MEC.

III. NONLINEAR BEHAVIOR

To solve the proposed MEC, Kirchhoff's voltage law is applied to each loop, generating the system of nonlinear equations reported, in (16) in its matrix form

$$\mathbf{A}_r \cdot \boldsymbol{\Phi} = \mathbf{F}. \quad (16)$$

Here, $\mathbf{A}_r$ represents the reluctance matrix associated with the network, $\boldsymbol{\Phi}$ is the magnetic flux vector and $\mathbf{F}$ denotes the vector of all magnetomotive force sources in the network.

The matrix $\mathbf{A}_r$ is constructed using the previously computed reluctances and is organized according to the network topology. A division into submatrices is shown in (17), where the diagonal matrices $\mathbf{A}_s$, $\mathbf{A}_{gap}$, $\mathbf{A}_{PM}$, and $\mathbf{A}_{rot}$ correspond to the reluctances of the stator, air gap, PMs (including leakage paths), and rotor, respectively. The off-diagonal terms represent the interactions between these machine parts

$$\mathbf{A}_r = \begin{bmatrix} \mathbf{A}_s & \mathbf{A}_{s,gap} & \mathbf{A}_{s,PM} & 0 \\ \mathbf{A}'_{s,gap} & \mathbf{A}_{gap} & \mathbf{A}_{gap,PM} & \mathbf{A}_{gap,rot} \\ \mathbf{A}'_{s,PM} & \mathbf{A}'_{gp,PM} & \mathbf{A}_{PM} & \mathbf{A}_{PM,rot} \\ 0 & \mathbf{A}_{gap,rot} & \mathbf{A}'_{PM,rot} & \mathbf{A}_{rot} \end{bmatrix}. \quad (17)$$

The matrix is populated following circuit theory principles for mesh-based network solutions. Specifically, the generic element on the main diagonal at position (i, i) represents the sum of all the reluctances in the i th loop. Conversely, the generic off-diagonal element at position (i, j) is the negative of the reluctances shared between the i th and j th loops.

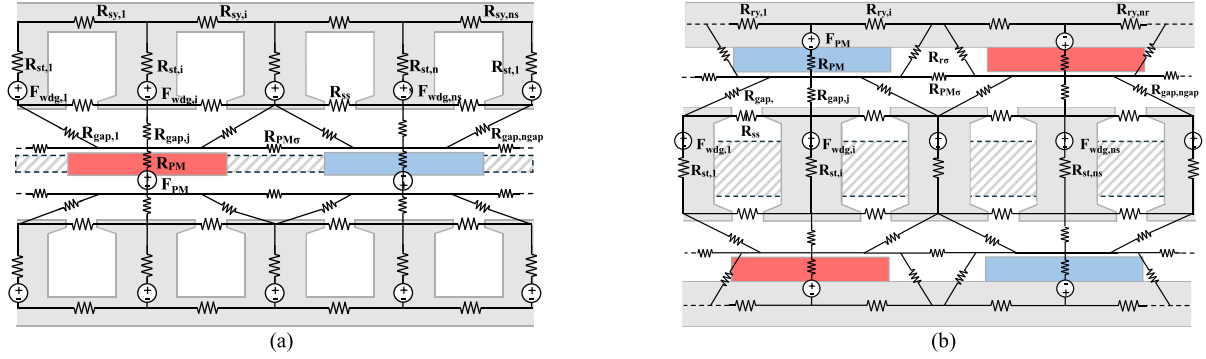


FIGURE 5. Magnetic equivalent circuit for (a) 2S1R-NN and (b) 1S2R-NS.

However, special attention must be given to the off-diagonal elements that include border reluctances, where symmetry conditions must be applied. Specifically, if the machine periodicity is odd, the sign of border reluctance must be reversed.

The vector F in (16) represents the MMF associated with each magnetic flux mesh. In the developed MEC, there are two sources of magnetomotive force: the stator windings and the PMs. For the PMs, the MMF is calculated using (12) and assigned to the corresponding loops. In contrast, the MMF contribution from the windings is determined using (18), which accounts for the total magnetomotive force generated by the windings

$$F_{wdg} = i_{abc} \cdot N_{wdg}. \quad (18)$$

In (18), F_{wdg} is the MMF vector associated with the winding, i_{abc} is the vector containing the instantaneous value of currents and N_{wdg} is the winding matrix that defines the winding arrangement within the slots with the specified number of turns.

Finally, the vector ϕ containing the magnetic fluxes associated with each loop and represents the unknown variable of the system. It is important to note that the loop fluxes do not always correspond directly to the actual fluxes in the circuit. For example, the reluctances in the stator yoke are effectively traversed by the stator loop fluxes, whereas, in the stator teeth, the actual flux is the difference between the fluxes of the adjacent loops, as in (3).

A. NUMERICAL SOLUTION PROCEDURE

The solution of the MEC is numerically obtained using the conjugate gradient (CG) solver [28], which is an iterative algorithm suited for large, sparse and symmetric problems of the form $A \cdot x = b$. The complete procedure is illustrated in Fig. 6. The method starts with an initial guess of the unknown loop flux vector Φ_0 . This initial guess is obtained by: calculating the network reluctances under unsaturated conditions (i.e., assuming maximum magnetic permeability); populating the reluctance matrix A_r ; and applying the inverse of (16).

The loop flux vector is subsequently used to calculate the flux in each circuit branch and, consequently, the magnetic

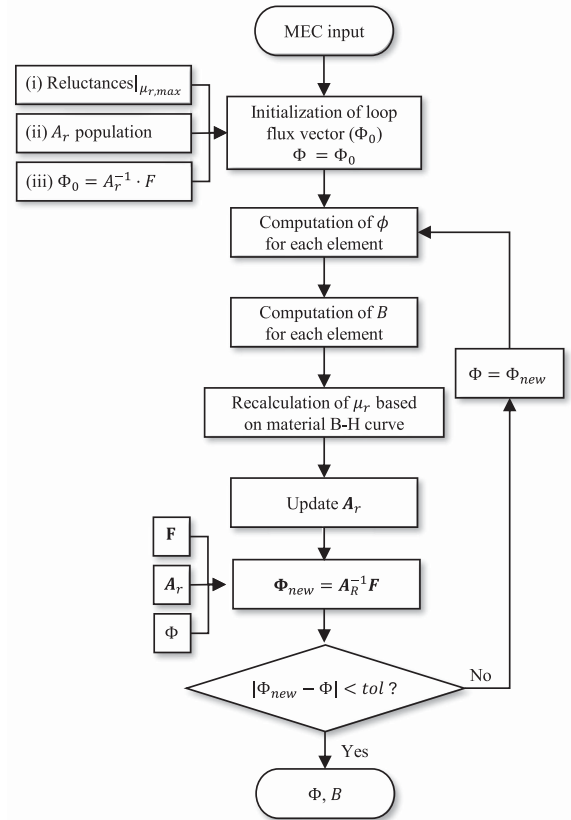


FIGURE 6. Procedure workflow.

flux density associated with each discretized network element. The obtained values of magnetic flux density allow for the computation of the corresponding magnetic permeability, thanks to the BH curve of the material. With the updated values of the magnetic permeabilities, the reluctances are recomputed and thus the reluctance matrix A_r is updated. Therefore, the CG solver is used to reach the convergence of the procedure.

Specifically, the solver computes the searching direction in such a way that generates the greatest improvement for each step. This method starts with the initial guess Φ_0 to compute

the associated initial residual r_0 as in (19)

$$r_0 = F - A_r \Phi_0. \quad (19)$$

The initial search direction p_0 is defined to coincide with r_0 and then, the successive residuals and search directions are computed iteratively to find the solution at each step

$$x_{k+1} = x_k + \left(\frac{p_k^T r_k}{r_k^T A_r p_k} \right) p_k. \quad (20)$$

In (20), the subscript k defines the current step during the iterative computation. Finally, once a convergence tolerance is reached, the process is stopped.

B. MECHANICAL ROTATION

The proposed MEC accounts for mechanical rotation by reevaluating the reluctance network at each rotor step, requiring the procedure outlined in Fig. 6 to be repeated continuously for each rotor position considered. To achieve this, the machine is modeled as an infinitely long stator, represented by repeating the slot pattern multiple times to emulate continuous rotation. Appropriate boundary conditions, such as periodicity, are applied to properly replicate the behavior of the magnetic field within the domain. The procedure determines, at each instant, how the stator teeth and air gap sections are connected, based on their relative positions. Using this information, the mesh loops are reassigned, and the reluctance matrix is rebuilt.

This approach allows the algorithm to handle any slot/pole combination, winding arrangement, and slot shape, as it relies solely on the relative positioning between the stator teeth and PMs. Additionally, this methodology enables the user to determine the extent to which each section faces a stator tooth, which is particularly useful for fringe flux calculations. Finally, it is worth noting that a relatively high discretization is recommended to accurately capture the magnetic interaction between each magnet portion and the stator iron.

IV. POSTPROCESS COMPUTATIONS

A. NO-LOAD AIR GAP FLUX DENSITY AND VOLTAGE

The MEC can be used to simulate the machine in no-load conditions. For a fixed rotor position, the air gap flux density can be obtained dividing the flux in each air gap element by the surface of the corresponding flux tube. The no-load magnetic flux density fundamental can be thus used to get the fundamental of the no-load voltage as in

$$E_1 = \frac{2\pi}{\sqrt{2}} \cdot f \cdot N_t \cdot k_{ew1} \cdot (\alpha_{PM} \cdot B_{g1} \cdot \tau_p \cdot L_{PM}) \quad (21)$$

where N_t is the number of turns in series per phase, and k_{ew1} is the fundamental winding factor.

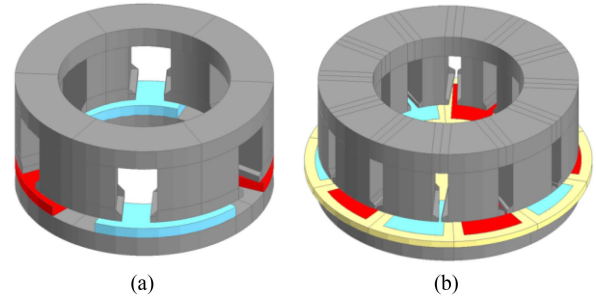


FIGURE 7. 3D sketches of the study cases. (a) M1. (b) M2.

TABLE 1. Main Specifications for the Study Cases

	M1	M2
Number of slots	6	12
Number of poles	4	10
Stator outer diameter, (mm)	180 mm	286 mm
Stator inner diameter, (mm)	120 mm	155 mm
Air gap thickness, (mm)	1 mm	6 mm
PM remanence, (T)	0.8 T	1.05 T

B. ELECTROMAGNETIC TORQUE

The analytical computation of the electromagnetic torque is based on the dq -axes formulation given as

$$T_{em} = \frac{3}{2} \cdot p \cdot (\lambda_d i_q - \lambda_q i_d) \quad (22)$$

where T_{em} is the electromagnetic torque, λ_d and λ_q are the flux linkages in the d - and q -axis, respectively. Similarly, i_d and i_q are the dq -axes currents, obtained from the stator phase currents using reference frame transformation matrices. The flux linkages in the windings can be calculated using

$$\lambda_{abc} = p \cdot N_{wdg} \cdot \Phi_{sy}. \quad (23)$$

Here, λ_{abc} is the linked flux in the abc reference frame and Φ_{sy} is the stator yoke flux vector.

V. MODEL VALIDATION

In order to validate the proposed reluctance network, a comparison with both finite element simulations and experimental tests is carried out.

A. FEA VALIDATION

For the FEA validation, two different single-stage axial-flux motor models with different pole/slot combinations and periodicities are considered. Fig. 7 shows the 3D sketches of the study cases (M1 and M2, respectively), while Table 1 lists their main dimensions. The results in terms of no-load magnetic flux density in the air gap, back electromotive force and electromagnetic torque under load conditions are evaluated. For the MEC, a fine air gap discretization is selected with at least 50 elements per pole.

Fig. 8 shows the resulting air gap magnetic flux density, obtained using both the proposed MEC and 3D FEA simulations. Notably, the largest discrepancies occur in regions where the

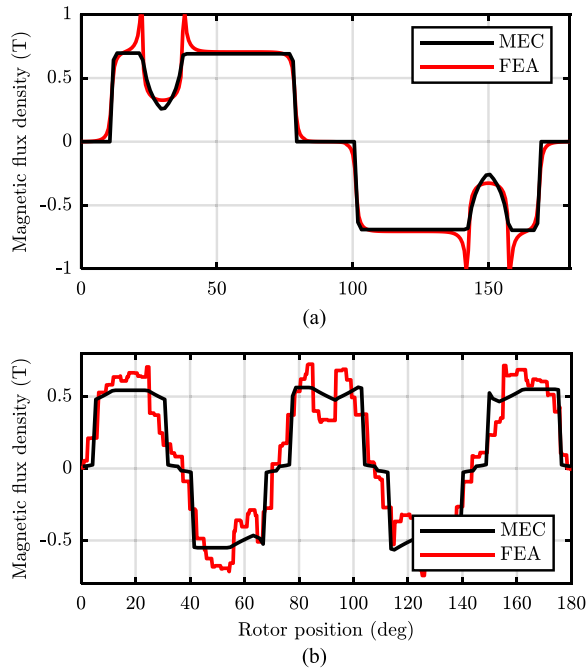


FIGURE 8. Air gap flux density comparison between MEC and FEA for (a) M1 and (b) M2.

TABLE 2. No-Load and Load Results Comparison for the Analyzed Study Cases

	M1			M2		
	MEC	FEA	$\Delta\%$	MEC	FEA	$\Delta\%$
B_g	0.65 T	0.66 T	1.5 %	0.61T	0.61 T	0.1 %
E_1	129.3 V	130.9 V	1.2%	107.8V	107.8 V	0.1%
T_{em}	24.7 N·m	25.0 N·m	1.1 %	40.6 N·m	40.3 N·m	1 %

slotting effect is more pronounced, reaching differences up to 10%, primarily due to the analytical computation of fringing flux, which is based on assumptions regarding the length of the flux lines entering the sides of the teeth (see Fig. 3, cases B and C). The fundamental value of the MEC magnetic flux density in the air gap is thus used to compute the fundamental value of the back electromotive force, as in (21). The comparison with the fundamental component of the FEA simulated back electromotive force is given in Table 2, resulting in an average error of less than 1.5% for both cases.

Fig. 9 compares the MEC and FEA models under load conditions in terms of torque values for both motors. While the MEC predictions of torque ripple relatively differ from those of the 3D FEA model—primarily due to the limited number of airgap reluctance elements considered in the network—the mean torque values obtained from both methods show good agreement. This is evident from the values reported in Table 2, which indicate that the maximum error in the average electromagnetic torque for the two study cases is in the order of 1%. These results confirm that the developed MEC is suitable for integration into fast design and optimization procedures. Accordingly, the model's accuracy is considered satisfactory

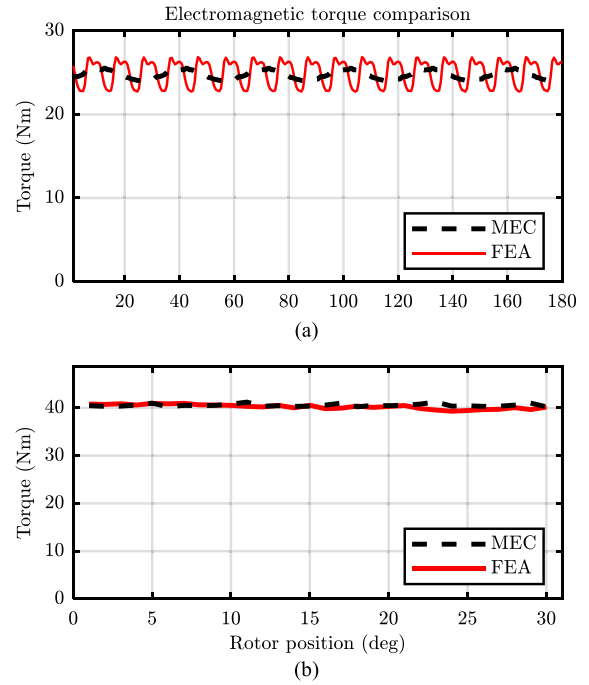


FIGURE 9. Electromagnetic torque comparison between MEC and FEA for (a) M1 and (b) M2.

when the mean values (for the torque) and the fundamental values (for emf and airgap flux density) show good agreement with the validation results. One limitation of the proposed MEC model, however, lies in its limited ability to accurately capture secondary effects such as torque ripple and airgap flux density harmonics. Although these phenomena are typically analyzed using FEA tools within the design workflow of electrical machines, they could potentially be addressed in the MEC framework by increasing the number of airgap reluctance elements per pole. An additional comparison under high-saturation conditions was performed for M2 to evaluate the performance of the procedure in overload scenarios. In this case, an operating point with more than twice the rated current was simulated. The results show a deviation of 2%, indicating that the MEC can still maintain high accuracy under these conditions.

It is worth noting that the results based on the MEC have been obtained with a significant reduction in computation time compared to FEA. Specifically, for the M2 case study, the computational time decreases from 5270 s for the FEA to 13 s for the MEC, making the proposed approach about 400 times faster. Both simulations were performed on the same workstation equipped with 128 GB of RAM and 20 CPU cores.

B. EXPERIMENTAL VALIDATION

To further validate the reluctance network, the results from the MEC are compared with experimental data obtained for an AFPM prototype. The specifications of the machine under test are the same as M2 in Table 1. Fig. 10 illustrates the

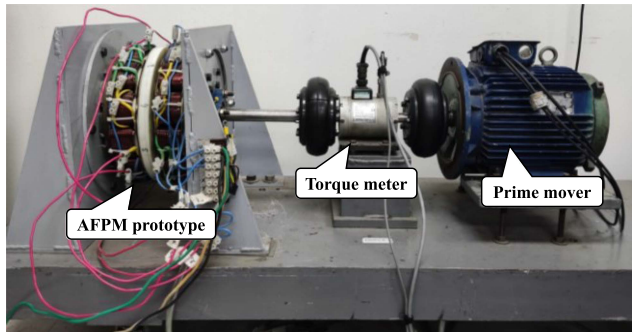


FIGURE 10. Experimental set-up for validation.

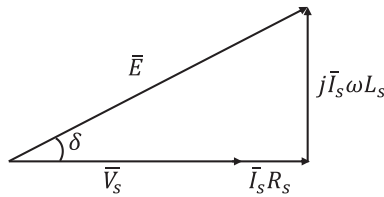


FIGURE 11. Phasor diagram of AFPM working as generator on a resistive load.

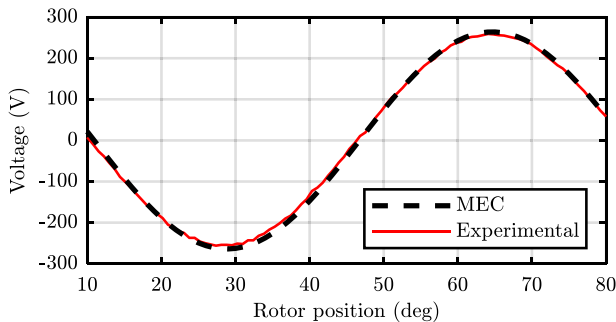


FIGURE 12. No-load voltage comparison.

set-up used for experimental testing, where the prototype has been operated as a generator connected to a resistive load. The machine under test has been driven by a prime mover and equipped with a torque meter.

To replicate the test conditions in the MEC, the load angle δ shown in the phasor diagram in Fig. 11 has to be computed. This requires the determination of the circuit parameters of the machine, i.e., the phase resistance R_s and the phase synchronous inductance L_s . In the phasor diagram, $\bar{V}_s$ is the terminal voltage, $\bar{E}$ is the open circuit electromotive force that is aligned with the q -axis, $\bar{I}_s$ is the phase current, and δ is the unknown load angle. The phase inductance has been obtained following the test procedure presented in [29], while a simple dc test allowed for the phase resistance measurement. Hence from the phasor diagram, the load angle has been computed as in (24). The obtained parameters and load angle are given

TABLE 3. Winding Parameters and Load Angle

Phase resistance (R_s)	0.18 (Ω)
Synchronous inductance (L_s)	13.60 (mH)
Load angle (δ)	11.17 (deg)

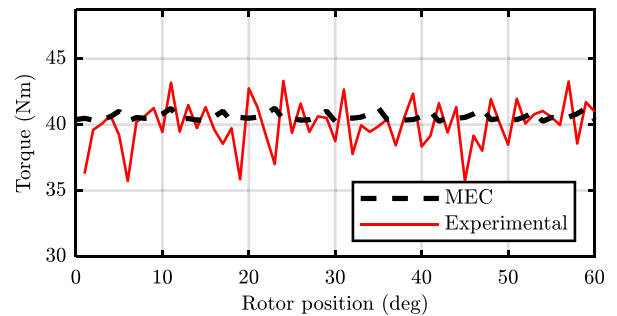


FIGURE 13. Torque comparison.

in Table 3

$$\delta = \arctan \left(\frac{I_s \omega L_s}{V_s + I_s R_s} \right). \quad (24)$$

For what it concerns the materials used in the MEC modeling, the datasheet BH curve of the prototype lamination has been implemented for the stator and rotor cores. Meanwhile, the remanence of the PMs has been determined through the calibration with the no-load voltage measurements. Fig. 12 shows the measured and calibrated MEC no-load voltage. Therefore, the resulting network has been used to predict the torque corresponding to the load angle of the experimental test. Fig. 13 shows the comparison of the test and MEC torque, revealing minor differences with an error on the average values below 3%.

VI. CONCLUSION

This article presents an MEC for the analysis of axial flux PM synchronous machines. The proposed methodology effectively integrates nonlinear iron behavior, rotor motion, and leakage effects enabling accurate predictions of key performance variables such as magnetic flux density, back electromotive force and electromagnetic torque. The proposed methodology is able to adapt to different axial flux machine configurations, ranging from the single-stage machine to double-stage arrangements with multiple stators or rotors.

The MEC model has been validated with both finite element simulations and experimental measurements, demonstrating strong agreement in the results. Compared to FEA, the proposed method significantly reduces simulation time, while maintaining accuracy in predicting the machine performance.

Given the good correlation with experimental results and its computational efficiency and adaptability, the proposed MEC represents a valuable tool for fast and preliminary axial flux machine design. Thanks to the analytical nature of the

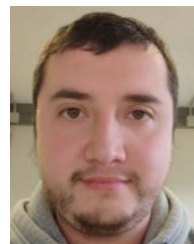
proposed model, it can be seamlessly integrated into optimization algorithms, sizing tools, and multiphysics analysis frameworks, enabling a more automated approach to machine design and performance evaluation.

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